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B.Tech. Degree V Semester Supplementary Examination June 2024

ME 19-205-0503 MACHINING SCIENCE AND MACHINE TOOLS (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: To study the geometry of single point and multi point cutting tools and mechanics of chip formation.
 CO2: To get the concept of mechanics of metal cutting.
 CO3: To understand the principles of working, specifications, different types and operations performed in lathe, drilling and shaping machines.
 CO4: To learn the principles of working, specifications, different types and operations performed in grinding and milling machines.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer *ALL* questions)

	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a)	Explain why negative rake angle is employed for cutting hard and strong materials.	3	L2	1	1
(b)	Explain how the American Standard Association (ASA) designation of tool nomenclature specify the geometry of a single point turning tool.	3	L1	3	1
(c)	Write a brief note on linear motion elements used in machine tools.	3	L1	3	1
(d)	How do the power transmission in machine tools is utilised to obtain a range of speeds and feeds required to cater various applications.	3	L2	3	2
(e)	If for turning a job, the length of the work is 200 mm and the taper 1:50. Find the set over required for the tail stock to achieve this taper.	3	L3	3	1
(f)	Differentiate between shaper and planer.	3	L2	3	1
(g)	Differentiate between plain milling and face milling.	3	L2	4	1
(h)	What is the standard method to specify a grinding wheel?	3	L2	4	1

PART B

(4 × 12 = 48)

II.	(a)	With a neat sketch explain the geometry of a single point cutting tool.	7	L2	2	1
	(b)	Discuss the thermal aspects of machining.	5	L2	2	2
OR						
III.	(a)	Explain how the tool wear affects the metal cutting performance.	6	L1	2	1
	(b)	In a metal cutting experiment, the following observations were made: rake angle = 10° Horizontal cutting force component = 1290 N Vertical component of cutting = 1650 N Calculate the various components of cutting forces and the coefficient of friction at the chip tool interface.	6	L3	2	2
IV.	(a)	Explain a method available for generating plane surfaces in machine tools.	6	L1	3	2
	(b)	How do the rotary motion is converted to linear motion in a machine tool? Explain.	6	L2	3	2
OR						
V.	(a)	Explain the kinematics of a shaper with the help of suitable sketches.	6	L2	3	2
	(b)	What are the functions served by a lead screw in a lathe? Give sketches.	6	L2	3	2

(P.T.O.)

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		Marks	BL	CO	PO
VI.	(a) Differentiate between capstan and turret lathe.	6	L2	3	1
	(b) Write a note on:	6	L1	3	2
	(i) universal chuck.				
	(ii) steady rest.				
OR					
VII.	(a) With a neat sketch explain the hydraulic quick return mechanism of shaper.	7	L1	3	1
	(b) Find the time required on the shaper to complete one cut on a plate 610 mm × 870 mm. The cutting speed is 6 m/min. The ratio of return time to cutting time is 1:5. Feed is 3 mm/stroke. The clearance at each end is assumed as 75 mm.	5	L3	3	2
VIII.	(a) Explain the compound indexing method in milling.	5	L1	4	1
	(b) A 300 mm × 50 mm rectangular cast iron piece is to be face milled with a carbide cutter. The cutting speed and feed are 50 m/min and 50 mm/min. If the cutter diameter is 80 mm and it has 12 cutting teeth, determine	7	L3	4	2
	(i) cutter r. p. m.				
	(ii) feed per tooth.				
	(iii) milling time.				
OR					
IX.	(a) Explain the 3-2-1 principle of location.	6	L2	4	1
	(b) What are the design and operational factors to be considered for proper and adequate clamping of a workpiece.	6	L2	4	1

Blooms's Taxonomy Level's

L1 – 30%, L2 – 52.5%, L3 – 17.5%.

